PRAKTIKUM SISTEM OPERASI

MODUL V

Eksplorasi Proses Xinu

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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JURNAL

1. **[10 Poin]** Jawablah pertanyaan berikut ini:
   a. Berapa banyaknya maksimum proses yang ada pada Xinu?
   8 Proses

```
xinu@xinu-develop-end:~/xinu$ cat ~/xinu/include/process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC           8
```

   b. Berapa maksimal panjang nama suatu proses pada Xinu?
   16 karakter

```
#define PNMLEN          16      /* Length of process "name"            */
```

   c. Berapa nilai prioritas awal pada saat proses dibuat?
   20

```
#define INITPRIO        20      /* Initial process priority            */
```

   d. Ada berapa total state pada Xinu? Sebutkan!
   8 Buah State

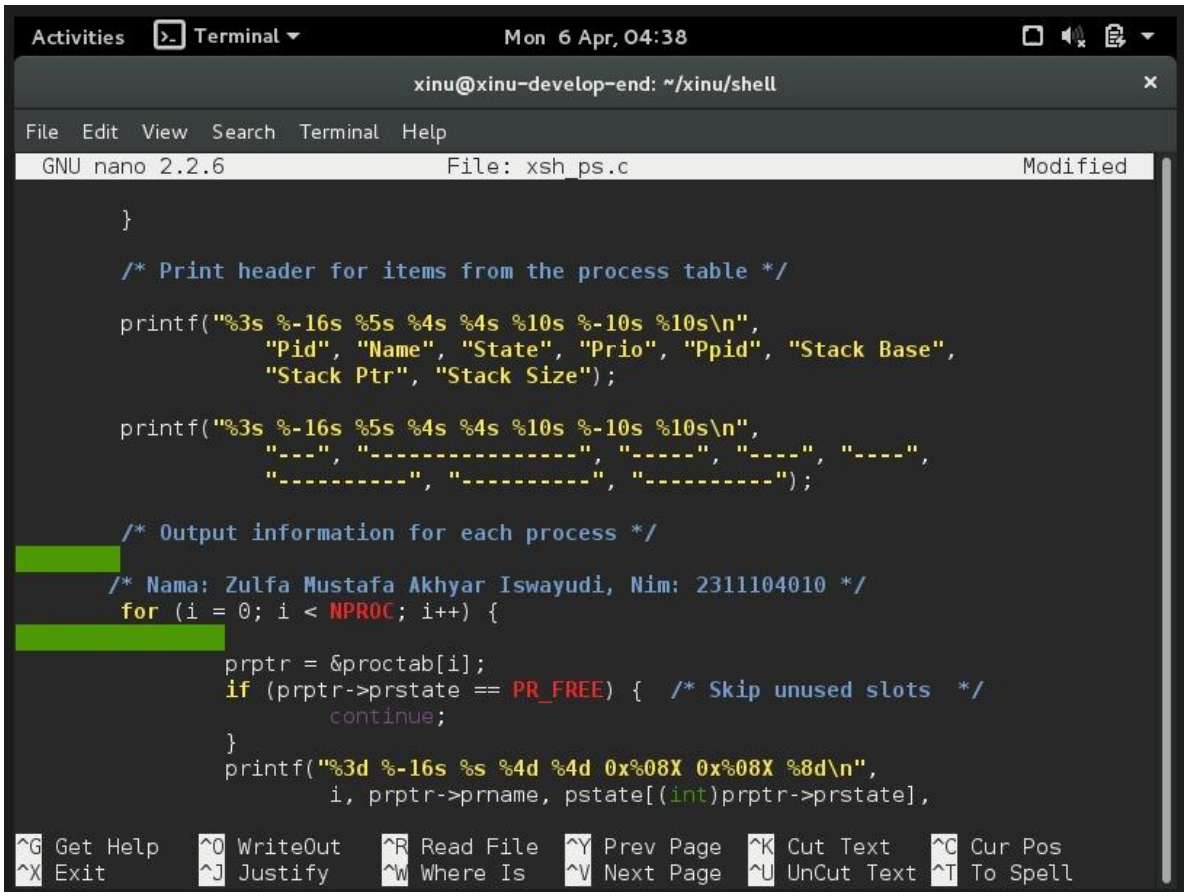
```
/* Process state constants */

#define PR_FREE         0       /* Process table entry is unused       */
#define PR_CURR         1       /* Process is currently running        */
#define PR_READY        2       /* Process is on ready queue           */
#define PR_RECV         3       /* Process waiting for message         */
#define PR_SLEEP        4       /* Process is sleeping                 */
#define PR_SUSP         5       /* Process is suspended                */
#define PR_WAIT         6       /* Process is on semaphore queue       */
#define PR_RECTIM       7       /* Process is receiving with timeout   */
```

2. **[20 Poin]** Perintah ps adalah perintah untuk menampilkan statistik process yang berjalan. Source code dari ps tersimpan pada file xsh_ps.c. Carilah file tersebut dan beri komentar pada 20 baris terakhir di source code tersebut!

```
xinu@xinu-develop-end:~/xinu$ find . -name "xsh_ps.c"
./shell/xsh_ps.c
xinu@xinu-develop-end:~/xinu$ cat shell/xsh_ps.c
/* xsh_ps.c - xsh_ps */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*------------------------------------------------------------------------
 * xsh_ps - shell command to print the process table
 *------------------------------------------------------------------------
 */
shellcmd xsh_ps(int nargs, char *args[])
{
        struct  procent *prptr;         /* Pointer to process           */
        int32   i;                      /* Index into proctabl          */
        char *pstate[]  = {             /* Names for process states     */
                "free ", "curr ", "ready", "recv ", "sleep", "susp ",
                "wait ", "rtime"};

        /* For argument '--help', emit help about the 'ps' command      */

        if (nargs == 2 && strncmp(args[1], "--help", 7) == 0) {
                printf("Use: %s\n\n", args[0]);
```

```
	/* Check for valid number of arguments */

	if (nargs > 1) {
		fprintf(stderr, "%s: too many arguments\n", args[0]);
		fprintf(stderr, "Try '%s --help' for more information\n",
				args[0]);
		return 1;
	}

	/* Print header for items from the process table */

	printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
		   "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
		   "Stack Ptr", "Stack Size");

	printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
		   "---", "----------------", "-----", "----", "----",
		   "----------", "----------", "----------");

	/* Output information for each process */

	for (i = 0; i < NPROC; i++) {
		prptr = &proctab[i];
		if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */
			continue;
```

```
	/* Output information for each process */

	for (i = 0; i < NPROC; i++) {
		prptr = &proctab[i];
		if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */
			continue;
		}
		printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
			i, prptr->prname, pstate[(int)prptr->prstate],
			prptr->prprio, prptr->prparent, prptr->prstkbase,
			prptr->prstkptr, prptr->prstklen);
	}
	return 0;
}
```



Pada proses ini kita mencoba menambahkan sedikit komentar yaitu nama dan nim didalam file xsh_ps.c. **File ini bertujuan untuk merubah hasil output tampilan tabel ps agar ditambahkan kolom baru.**

3. **[35 Poin]** Ubahlah perintah ps (source code: xsh_ps.c) pada Xinu sehingga menampilkan informasi tambahan berupa kolom yang berisi total message yang ada pada proses. Jawaban:

```
Activities  Terminal ▾        Mon 6 Apr, 01:21
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Tabs  Help
xinu@xinu-develop-end: ~/xinu      xinu@xinu-develop-end: ~/xinu
GNU nano 2.2.6          File: shell/xsh_ps.c                Modified

            fprintf(stderr, "Try '%s --help' for more information\n",
                            args[0]);
            return 1;
        }

        /* Print header for items from the process table */

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
                   "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
                   "Stack Ptr", "Stack Size", "Msg", "Content");

        printf("%-4s %-16s %-6s %-5s %-4s %10s %-10s %-10s %-3s, %-7s\n",
                   "---", "----------------", "-----", "----", "----",
                   "----------", "----------", "----------", "---", "-------");

        /* Output information for each process */

        for (i = 0; i < NPROC; i++) {
                prptr = &proctab[i];
                if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */

^G Get Help   ^O WriteOut   ^R Read File   ^Y Prev Page   ^K Cut Text   ^C Cur Pos
^X Exit       ^J Justify    ^W Where Is    ^V Next Page   ^U UnCut Text ^T To Spell
```

Ini adalah baris dimana kita menambah kolom-kolom baru beserta placeholder dan jarak antar kolom yang contoh formatnya **%5s**.

```
/* Output information for each process */

for (i = 0; i < NPROC; i++) {
        prptr = &proctab[i];
        if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */
                continue;
        }
        printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d %3d %7d\n",
                i, prptr->prname, pstate[(int)prptr->prstate],
                prptr->prprio, prptr->prparent, prptr->prstkbase,
                prptr->prstkptr, prptr->prstklen, prptr->prhasmsg, prptr->prms$
}
return 0;
```

Ini adalah contoh kita menambahkan data pemrosesan kedalam loop kompilasi sistem.

4. **[35 Poin]** Ubahlah perintah uptime pada Xinu sehingga menampilkan lamanya Xinu sejak booting hanya dalam satuan menit:

```
GNU nano 2.2.6                File: shell/xsh_uptime.c

/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*------------------------------------------------------------------------
 * xsh_uptime - shell to print the time the system has been up
 *------------------------------------------------------------------------
 */
shellcmd xsh_uptime(int nargs, char *args[])
{
        uint32  days, hrs, mins, secs;  /* Days, hours, minutes,  and   */
                                        /*  seconds since system boot   */
        uint32  secperday = 86400;      /* Seconds in a day             */
        uint32  secperhr  =  3600;      /* Seconds in an hour           */
        uint32  secpermin =    60;      /* Seconds in a minute          */

        /* For argument '--help', emit help about the 'uptime' command  */

                                [ Read 75 lines ]
^G Get Help   ^O WriteOut   ^R Read File  ^Y Prev Page  ^K Cut Text   ^C Cur Pos
^X Exit       ^J Justify    ^W Where Is   ^V Next Page  ^U UnCut Text ^T To Spell
```

Masuk ke folder xinu/shell dan buka xsh_uptime.c. Muncul output seperti ss-an diatas.

```
GNU nano 2.2.6            File: shell/xsh_uptime.c

                fprintf(stderr, "Try '%s --help' for more information\n",
                                args[0]);
                return 1;
        }

        secs = clktime;         /* Total seconds since boot */

        /* Hitung total menit sejak boot */
        mins = clktime / secpermin;
        printf("Xinu has been up %d minute(s)\n", mins);

        return 0;
}
```

Percabangan date time disederhanakan menjadi format satuan menit saat kita menjalankan perintah uptime di minicom nanti.

```
----------------------------------------

  __   __   ____   _   _   _   _
  \ \ / /  |_ _|  | \ | |  | |  | |
   \ V /    | |   |  \| |  | |  | |
   / /\ \   _| |_  | \  |  | |  | |
  / /  \ \  |    | | | \ |  \ --  /
  --    --  -----   -   -    ----
----------------------------------------


Welcome to Xinu!


xsh $ ps
Pid Name            State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
--- --------------- ----- ---- ---- ---------- ---------- ---------- ---, -------
  0 prnull          ready    0    0 0x005FDFFC 0x00FFFEE4     8192   1       3
  1 ipout           wait   500    0 0x005FBFFC 0x005FBF30     8192   0       0
  2 netin           wait   500    0 0x005F9FFC 0x005F9ED0     8192   0       0
  4 Main process    recv    20    3 0x005E7FFC 0x005E7F34    65536   0       0
  5 shell           recv    50    4 0x005D7FFC 0x005D79AC     8192   0       0
  6 ps              curr    20    5 0x005F7FFC 0x005F7FC4     8192   0       0
xsh $ uptime
Xinu has been up 15 minute(s)
```

Hasil akhir **berupa tabel ps yang ada dua kolom baru dengan proses datanya.** Sedangkan untuk uptime satuan waktu yang tampil adalah **15 minute(s)**